Proposed Resolution for NB Comment GB13-309 `atomic_ref<T>` is not convertible to `atomic_ref<const T>`

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Reply-to:	Hui Xie <hui.xie1990@gmail.com> Damien Lebrun-Grandié <lebrungrandt@ornl.gov>

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§ 1 Abstract

This paper proposes a converting constructor for `atomic_ref` as the resolution for UK NB Comment GB13-309.

§ 2 Introduction

The UK NB Comment GB13-309 pointed out that `atomic_ref<T>` is not convertible to `atomic_ref<const T>`. [P3323R1] added cv qualifiers to `atomic` and `atomic_ref`. However, the conversion constructor of `atomic_ref` between different cv qualifiers is overlooked.

§ 3 Wording

Change [atomics.ref.generic.general] as follows:

```
    constexpr explicit atomic_ref(T&);
    constexpr atomic_ref(const atomic_ref&) noexcept;
    template <class U>
    constexpr atomic_ref(const atomic_ref<U>&);
```

Add a new entry to [atomics.ref.ops] after paragraph 8:

```
template <class U>
constexpr atomic_ref(const atomic_ref<U>& ref);
```

9 *Contraints*:

(9.1) — `is_same_v<remove_cv_t<T>, remove_cv_t<U>>` is `true`, and

(9.2) — `is_convertible_v<U*, T*>` is `true`

10 *Postconditions*: `*this` references the object referenced by `ref`.

§ 4 Implementation Experience

[P3323R1] is currently being implemented in libc++ and adding the proposed constructor is under way.

§ 5 Feature Test Macro

[P3323R1] does not seem to mention Feature Test Macro

§ 6 References

[P3323R1] Gonzalo Brito Gadeschi. 2024-11-18. cv-qualified types in atomic and atomic_ref.
https://wg21.link/p3323r1